TryHackMe Writeup: What Is Networking?

- **Platform:** TryHackMe
- **Room:** https://tryhackme.com/room/whatisnetworking
- **Difficulty:** Introductory
- **Topics:** Networks, the internet, IP addresses, MAC addresses, MAC spoofing, and ping/ICMP

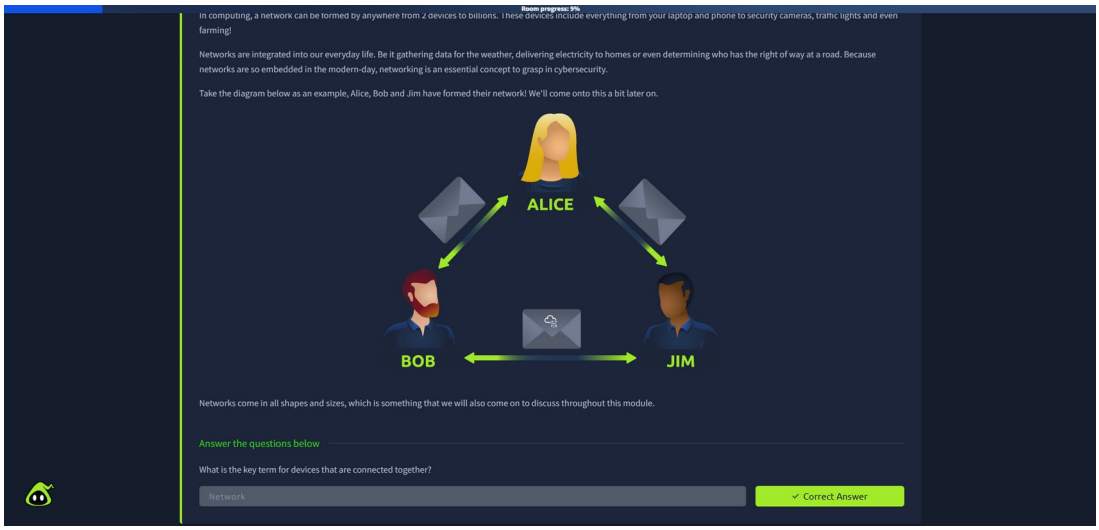
1. Objective

This room introduces basic computer-networking concepts and shows how devices identify and communicate with each other. It explains what a network is, introduces the World Wide Web, describes IP and MAC addresses, and demonstrates the ping utility and ICMP.

2. Task and Method

Task 1 - What Is Networking?

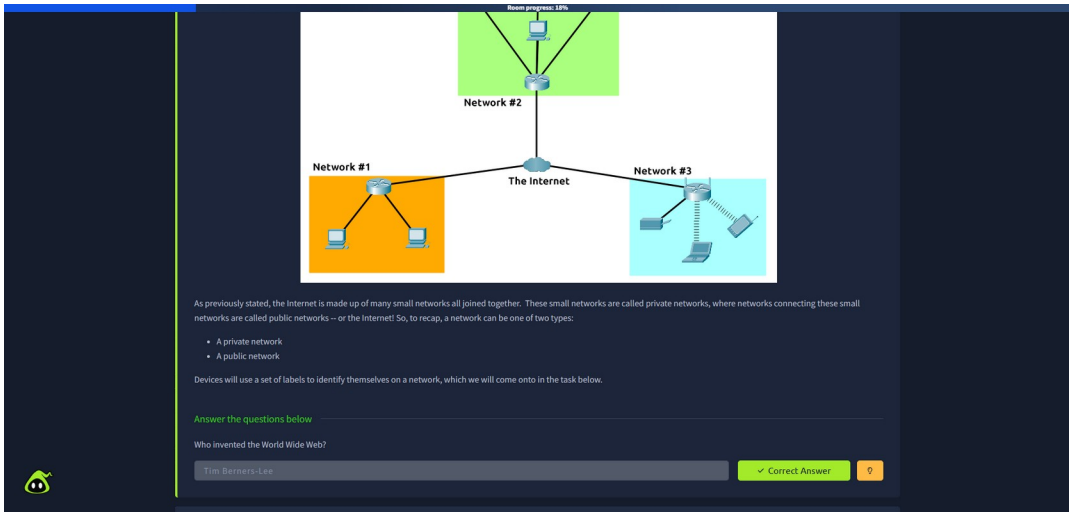
A **network** is a group of devices connected so they can communicate or share information and resources. Devices may include computers, phones, servers, printers, routers, and other smart devices.



Answer: The key term for devices that are connected together is Network.
Learning point: Networks may be small, such as a home Wi-Fi network, or large, such as the internet. Network protocols define how devices exchange information.

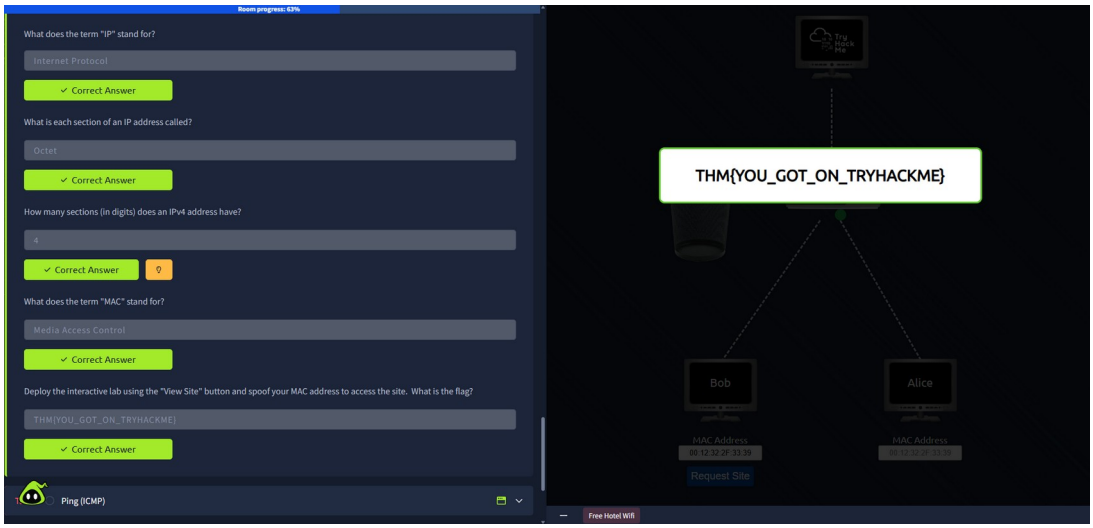
Task 2 - What Is the Internet?

The internet is a global network of interconnected networks that communicate using standard protocols. The **World Wide Web (WWW)** is a service that operates over the internet and allows people to access linked pages and resources through web browsers.



Answer: The inventor of the World Wide Web is Tim Berners-Lee.
Learning point: The internet and the World Wide Web are not identical. The internet is the underlying network infrastructure, while the Web is one of the services that uses it.

Task 3 - Identifying Devices on a Network
IP address
An **IP (Internet Protocol) address** is a logical address used to identify a network interface and route packets between networks. IPv4 addresses are commonly written as four decimal numbers separated by dots, for example 192.168.1.10. Each of the four numbers is called an **octet**, because it represents eight bits.
MAC address
A **MAC (Media Access Control) address** is a link-layer identifier used for local network communication. MAC addresses are commonly represented as six pairs of hexadecimal digits.



Answers:

- What does IP stand for? Internet Protocol
- What is each section of an IPv4 address called? Octet

- How many sections does an IPv4 address have? 4
- What does MAC stand for? Media Access Control

Interactive lab: MAC address spoofing

1. Click **View Site** in the TryHackMe task to open the provided simulation.
2. Review the scenario showing one device permitted to access the site and another device blocked.
3. In the simulation only, change the blocked device's MAC address to match the permitted device, following the lab instructions.
4. Observe the result and record the flag shown by the exercise.

Flag reported by a room walkthrough: THM{YOU_GOT_ON_TRYHACKME}

Security lesson: MAC address filtering alone is not strong authentication because MAC addresses can be spoofed. Real networks should use stronger access controls. Practise spoofing only in the authorized TryHackMe simulation or another network you own or have explicit permission to test.

Task 4 - Ping (ICMP)

ping is a command-line tool that sends network probes and reports responses, helping check reachability and estimate round-trip time. It commonly uses **ICMP (Internet Control Message Protocol)** for these tests.

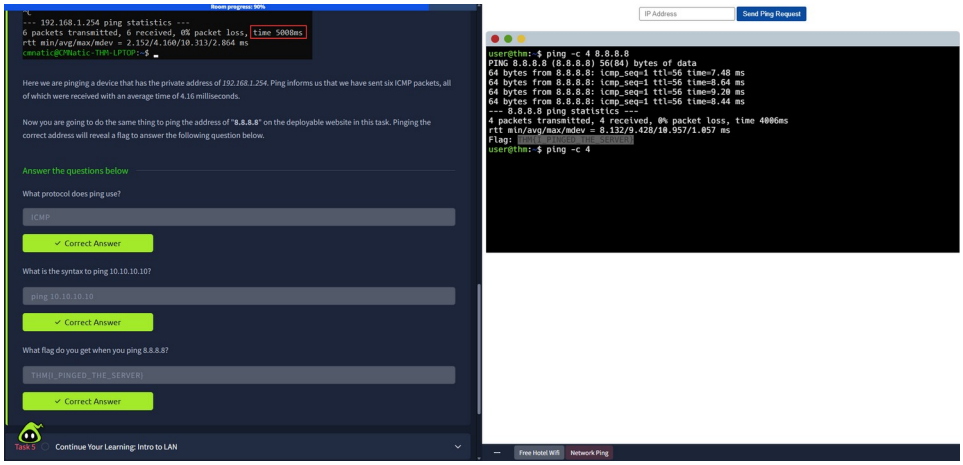
Check a specified IP address:

ping 10.10.10.10

Check a public DNS server:

ping 8.8.8.8

A destination may not reply if a firewall blocks ICMP, the route is unavailable, or the destination does not respond. Therefore, no ping response does not automatically mean the host is offline.



Answers:

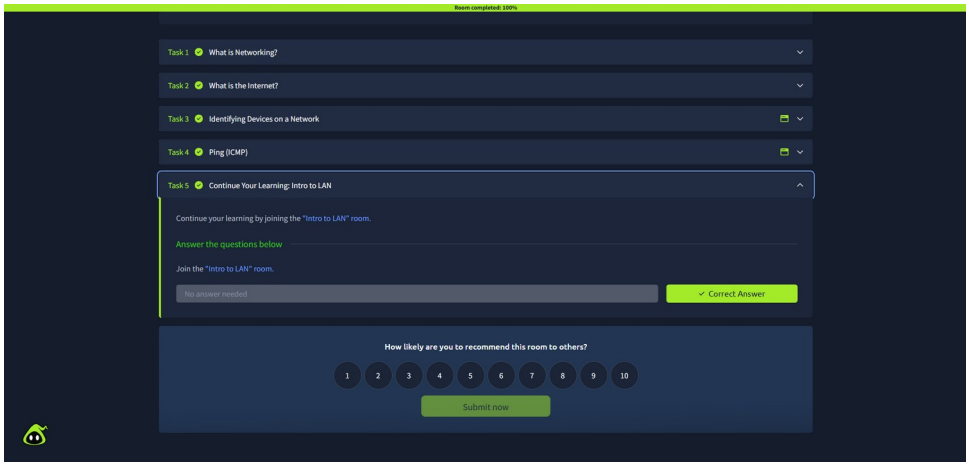
- What protocol does ping use? ICMP
- What is the syntax to ping 10.10.10.10? ping 10.10.10.10

- What flag is shown when pinging 8.8.8.8 in the lab? THM{I_PINGED_THE_SERVER}

Learning point: Ping is useful for basic connectivity troubleshooting, but it does not confirm that every application or port on a device is reachable.

Task 5 - Continue Your Learning

The room recommends continuing with **Intro to LAN** to learn about local-area networks and the technologies used to connect devices.



Completion note: Read the section and follow the room's completion instructions. Add a screenshot of the completed room progress.

3. Answer Summary

Task	Question	Answer
1	Key term for devices connected together	Network
2	Inventor of the World Wide Web	Tim Berners-Lee
3	Meaning of IP	Internet Protocol
3	Each IPv4 address section	Octet
3	Number of IPv4 sections	4
3	Meaning of MAC	Media Access Control
3	Flag from MAC address simulation	THM{YOU_GOT_ON_TRYHACKME}
4	Protocol used by ping	ICMP
4	Ping syntax for 10.10.10.10	ping 10.10.10.10
4	Flag from the 8.8.8.8 lab exercise	THM{I_PINGED_THE_SERVER}

The interactive room content may change. Confirm the flags and expected answers on the official TryHackMe page before submitting.

4. Key Concepts Learned

- A network connects devices so they can communicate and share resources.

- The internet is the global network of networks; the Web is a service that runs over it.
- IP addresses support logical addressing and routing.
- MAC addresses help deliver frames on a local network.
- MAC spoofing can bypass simple MAC-based filters, which is why MAC filtering should not be the only access-control method.
- ping commonly uses ICMP to test reachability and measure response time.
- A successful ping does not prove that a host's applications are secure or that a particular TCP/UDP port is open.